Internet and Web Programming

Module 1

Introduction to Web System

Internet Overview- WWW - Web Protocols - Web Browsers and Web Servers - Web System Architecture – URL - Domain Name – Client and Server-side Scripting.

Internet and Web:

- The internet is a global network of interconnected computers and servers that allows people to communicate, share information, and access resources from anywhere in the world. It was created in the 1960s by the US Department of Defense as a way to connect computers and share information between researchers and scientists.
- The World Wide Web, or simply the web, is a system of interconnected documents and resources, linked together by hyperlinks and URLs. It was created by Tim Berners-Lee in 1989 as a way for scientists to share information more easily. The web quickly grew to become the most popular way to access information on the internet.
- Together, the internet and the web have revolutionized the way we communicate, do business, and access information. They have made it possible for people all over the world to connect with each other instantly and have transformed many industries, from media and entertainment to education and healthcare.

Internet and Web:

- **The Internet:** In simplest terms, the Internet is a global network comprised of smaller networks that are interconnected using **standardized** communication protocols. The Internet standards describe a framework known as the Internet protocol suite. This model divides methods into a *layered system of protocols.*
- These layers are as follows:
  - **Application layer (highest) -** concerned with the data(URL, type, etc.). This is where HTTP, HTTPS, etc., comes in.
  - **Transport layer -** responsible for end-to-end communication over a network.
  - **Network layer -** provides data route.
- The Internet provides a variety of information and communication facilities; contains forums, databases, email, hypertext, etc. It consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.

Internet and Web:

- **The World Wide Web:** The Web is a major means of access information on the Internet. It's a system of Internet servers that support specially formatted documents.
- The documents are formatted in a markup language called **HTML**, or "HyperText Markup Language", which supports a number of features including links and multimedia. These documents are interlinked using hypertext links and are accessible via the Internet.
- To link hypertext to the Internet, we need:
    - The markup language, i.e., HTML.
    - The transfer protocol, e.g., HTTP.
    - Uniform Resource Locator (URL), the address of the resource.
- We access the Web using **Web browsers**.

Difference between Web and Internet:

Internet	Web
The Internet is the network of networks and the network allows to exchange of data between two or more computers.	The Web is a way to access information through the Internet.
It is also known as the Network of Networks.	The Web is a model for sharing information using the Internet.
The Internet is a way of transporting information between devices.	The protocol used by the web is HTTP.
Accessible in a variety of ways.	The Web is accessed by the Web Browser.
Network protocols are used to transport data.	Accesses documents and online sites through browsers.
Global network of networks	Collection of interconnected websites
Access Can be accessed using various devices	Accessed through a web browser

Difference between Web and Internet:

Connectivity Network of networks that allows devices to communicate and exchange data	Connectivity Allows users to access and view web pages, multimedia content, and other resources over the Internet
Protocols TCP/IP, FTP, SMTP, POP3, etc.	Protocols HTTP, HTTPS, FTP, SMTP, etc.
Infrastructure Consists of routers, switches, servers, and other networking hardware	Infrastructure Consists of web servers, web browsers, and other software and hardware
Used for communication, sharing of resources, and accessing information from around the world	Used for publishing and accessing web pages, multimedia content, and other resources on the Internet
No single creator	Creator Tim Berners-Lee
Provides the underlying infrastructure for the Web, email, and other online services	Provides a platform for publishing and accessing information and resources on the Internet

Evolution of Internet

Key Milestones in Internet History

- ***1969:** Launch of **ARPANET**, the first operational packet-switching network.*
- ***1971:** The first **email** is sent by **Ray Tomlinson**.*
- ***1983:** Adoption of **TCP/IP** as the standard protocol for ARPANET.*
- ***1989-1990: Tim Berners-Lee** invents the **World Wide Web**.*
- ***1993:** The first graphical web browser, **Mosaic**, is released.*
- ***1998: Google** is founded, revolutionizing search engines.*
- ***2007: The iPhone** introduces widespread mobile Internet access.*
- ***2010s:** Rise of **cloud computing**, **social media**, and **streaming services**.*

Evolution of Internet

1970s: The Foundation of Modern Networking

- The concept of Internet was originated in 1969 and has undergone several technological & Infrastructural changes as discussed below:
- The origin of Internet devised from the concept of **Advanced Research Project Agency Network (ARPANET).**
- **ARPANET** was developed by United States Department of Defence. Basic purpose of ARPANET was to provide communication among the various bodies of government.
- Initially, there were only four nodes, formally called Hosts. In 1972, the **ARPANET** spread over the globe with 23 nodes located at different countries and thus became known as **Internet.**
- **Birth of TCP/IP Protocol**: Cerf and Kahn's development of TCP/IP became the primary protocol suite for the Internet, establishing a standardized method for network communication in 1978.

Evolution of Internet

1980s: From Academic Network to Public Service

1983

- **Transition to TCP/IP**: On January 1, 1983, ARPANET switched entirely to TCP/IP, marking the official birth of the modern Internet as we know it. This event is celebrated as the beginning of all Internet communications.
- **The Domain Name System (DNS)**: Paul Mockapetris invented the DNS, introducing domains like .com, .org, and .edu, making the Internet more user-friendly.

1984

- The **Domain Name System (DNS)** was introduced, creating recognizable addresses such as ".com," ".org," and ".net." This simplified the way people accessed websites, making it more user-friendly.

1989

- **The World Wide Web is Conceived**: Tim Berners-Lee, a British scientist working at CERN, proposed an idea for a distributed information system that became the World Wide Web. His goal was to create a way to link and access documents over the Internet using hyperlinks.

Evolution of Internet

1990s: The Rise of the Web and Commercial Internet

1990

- **WWW Code Released**: Berners-Lee wrote the first web page editor and browser, and the WWW was officially born. It was the dawn of websites and web pages. Berners-Lee developed the first web browser and web server, bringing his idea of the **World Wide Web** to life. This marked the beginning of web browsing and laid the foundation for websites and content sharing.
- **ARPANET Shutdown**: ARPANET, which had become obsolete, was decommissioned, officially passing the torch to the modern Internet.

1991

- The **World Wide Web** was publicly released, enabling people to share and access documents and websites through a standardized system.

1993

- **Mosaic Browser**: Marc Andreessen and Eric Bina developed Mosaic, the first popular web browser that brought a graphical interface to the web. This innovation allowed non-technical people to explore the web easily.

1998

- **Google's Founding**: Larry Page and Sergey Brin launched Google, fundamentally changing how people accessed information. Its search algorithm quickly became the most efficient way to find information online.

Evolution of Internet

2000s: The Explosion of Social Media and Mobile Internet

2001

- **Wikipedia Launched**: A free, collaborative online encyclopedia, Wikipedia became one of the most visited websites on the Internet, transforming the way knowledge was shared.

2004

- **Facebook is Born**: Originally a social network for college students, Facebook now known as Meta (founded by Mark Zuckerberg) rapidly expanded to become one of the largest social platforms in the world.

2005

- **YouTube’s Founding**: The video-sharing platform YouTube transformed content creation and consumption, giving rise to video blogging (vlogging) and viral content.

2008

- The **Google Chrome** browser debuted, offering faster browsing speeds and setting new standards for web performance.

Evolution of Internet

2010s: The Age of Connectivity, Apps, and Social Influence

2013

- **Edward Snowden Revelations**: Leaks by former NSA contractor Edward Snowden exposed the extent of global digital surveillance, sparking worldwide debate about online privacy and government oversight.

2015

- **Internet of Things (IoT)**: The term gained popularity as devices from thermostats to cars became connected to the Internet, transforming everyday objects into "smart" devices.

2016

- **Artificial Intelligence and Internet Integration**: AI technologies, chatbots, and smart assistants like Amazon's Alexa and Google Assistant became part of everyday Internet usage.

2018

- **GDPR (General Data Protection Regulation)** was enforced in the European Union, highlighting privacy concerns and regulating how companies handle user data.

Evolution of Internet

2020s: The Era of Remote Work and AI Integration

2021

- The **Metaverse** concept gained traction, with major tech companies like **Meta (formerly Facebook)** exploring virtual reality spaces for socialization, work, and commerce.

2022

- **Web3 and Decentralization**: Conversations around Web3, blockchain, decentralized finance (DeFi), and cryptocurrencies captured global attention, promising a more decentralized Internet structure.

2024

- **AI Dominance and Internet Personalization**: AI continues to shape user experiences online. From advanced content recommendations to seamless digital assistants, personalization remains a key driver for the future of the Internet.
- Continued expansion of **5G** technology and discussions around **AI ethics** shape the modern Internet, influencing how connectivity is managed and regulated worldwide.

2025

- **Metaverse and Virtual Reality**
  By 2025, the Metaverse will gain more prominence, with virtual spaces for socializing, working, and shopping.
- **Ethical AI and Privacy Concerns**
  As AI becomes more integral to the internet, issues around its ethical use and user privacy will become more pressing

Advantages

- Internet allows us to communicate with the people sitting at remote locations. There are various apps available on the wed that uses Internet as a medium for communication. One can find various social networking sites such as: Facebook, Twitter, Yahoo
- One can surf for any kind of information over the internet. Information regarding various topics such as Technology, Health & Science, Social Studies, Geographical Information, Information Technology, Products etc can be surfed with help of a search engine.
- Apart from communication and source of information, internet also serves a medium for entertainment. Following are the various modes for entertainment over internet. Online Television, Online Games, Songs, Videos, Social Networking Apps
- Internet allows us to use many services like: Internet Banking, Matrimonial Services, Online Shopping, Online Ticket Booking, Online Bill Payment, Data Sharing, E-mail
- Internet provides concept of **electronic commerce**, that allows the business deals to be conducted on electronic systems

Disadvantages

- There are always chances to lose personal information such as name, address, credit card number. Therefore, one should be very careful while sharing such information. One should use credit cards only through authenticated sites.
- Another disadvantage is the **Spamming**. Spamming corresponds to the unwanted e-mails in bulk. These e-mails serve no purpose and lead to obstruction of entire system.
- **Virus** can easily be spread to the computers connected to internet. Such virus attacks may cause your system to crash or your important data may get deleted.
- Also, a biggest threat on internet is pornography. There are many pornographic sites that can be found, letting your children to use internet which indirectly affects the children healthy mental life.
- There are various websites that do not provide the authenticated information. This leads to misconception among many people.

World Wide Web (www)

- The World Wide Online (WWW), often known as the Web, is an information system in which Uniform Resource Locators (URLs) are used to identify documents and other digital resources. URLs, such as
- https://example.com/, which can be connected together via hyperlinks and are accessible over the Internet.
- Hypertext Transport Protocol (HTTP) is used to transport Web resources, which are accessed by users via a web browser and published by a web server. The World Wide Web is not the same as the Internet, which is built on the same technology as the Web and predates it by more than two decades in certain forms.

A Brief History of www:

- The World Wide Web was founded in 1989 by Tim Berners-Lee, an English scientist.
- He invented the first web browser in 1990 while working at CERN in Geneva, Switzerland.
- The browser was first made accessible to other research organizations outside of CERN in January 1991, and then to the general public in August 1991.
- In the years 1993–2004, when websites for general use were available, the Web began to become more widely used.
- The World Wide Web is the tool that billions of people use to interact on the Internet, and it has played a key role in the development of the Information Age.

Web Protocol:

Protocol:

- A **protocol** is a set of predefined rules that handle how data is exchanged between computers over the Internet. It ensures that devices can communicate in a structured, reliable, and secure manner. In essence, protocols dictate how data is sent, received, formatted, and processed.
- Commonly known as **web protocols**: **HTTP, HTTPS, TCP/IP, FTP, and DNS**

Web Protocol:

HTTP (Hyper Text Transfer Protocol)

- HTTP Protocol is used to transfer hypertexts over the internet and it is defined by the www(world wide web) for information transfer.
- HTTP is a communication protocol. It defines mechanism for communication between browser and the web server. It is also called request and response protocol because the communication between browser and server takes place in request and response pairs.
- However, HTTP is **not secure**, meaning data transmitted over HTTP can be intercepted by malicious actors. It is used to share text, image.

HTTP Request : HTTP request comprises of lines which contains:

- Request line
- Header Fields
- Message body

Web Protocol:

HTTP (Hyper Text Transfer Protocol)

Key Points

- The first line i.e. the **Request line** specifies the request method i.e. **Get** or **Post.**
- The second line specifies the header which indicates the domain name of the server from where index.htm is retrieved.

HTTP Response: Like HTTP request, HTTP response also has certain structure. HTTP response contains:

- Status line
- Headers
- Message body

- When a user enters a website URL, the client sends an HTTP request to the server asking for the webpage or specific data. The server then processes this request and sends back an HTTP response containing the required content. This entire exchange happens over a TCP (Transmission Control Protocol) connection, which ensures that the data is sent reliably and in the correct order. HTTP is the communication protocol that makes this client-server interaction possible, allowing users to access web content.

Web Protocol:

HTTPS:

- **HTTPS (HyperText Transfer Protocol Secure)** is the secure version of HTTP. It works the same way as HTTP by allowing your browser to request and receive web pages from a server, but with one important difference — it **encrypts** the data being exchanged.
- This means any information sent between your browser and the website (like passwords, credit card details, or personal data) is kept private and safe from hackers. HTTPS uses **SSL/TLS encryption** to protect the connection, making it the preferred and trusted protocol for secure websites, especially for online shopping, banking, or login pages.
- The secure communication process in HTTPS involves the following key steps:
- **TCP Connection**: First, a stable TCP connection is established between the client and the server to start communication.
- **Public Key**: The server sends its **public key** to the client. This key is part of an SSL certificate and is used to safely exchange information without exposing it to attackers.
- **Session Key**: The client then generates a **session key** and encrypts it using the server's public key. This ensures that only the server can decrypt and access the session key.
- **Data Encryption:** Once the session key is exchanged, all data transferred between the client and server is **encrypted**. This keeps personal information, passwords, and other sensitive data safe from hackers.

Web Protocol:

3. TCP (Transmission Control Protocol)

- TCP (Transmission Control Protocol)is a communication protocol that ensures reliable, ordered, and error-checked delivery of data between devices over a network. It breaks data into packets, sends them to the destination, and reassembles them in the correct order.
- **TCP (Transmission Control Protocol)** establishes a reliable connection between a **client** (like your computer) and a server using a method called the **three-way handshake**.
    - **SYN (Synchronize)**: The client starts the process by sending a **SYN** message to the server, asking to start a connection.
    - **SYN + ACK (Acknowledge)**: The server receives the SYN request and responds with a **SYN-ACK** message, which means it agrees to the connection and acknowledges the client's request.
    - **ACK (Acknowledge)**: The client replies with an **ACK** message, confirming the connection.
- TCP offers:
    - Stream Data Transfer.
    - Reliability.
    - Efficient Flow Control
    - Full-duplex operation.
    - Multiplexing.

Web Protocol:

3. TCP (Transmission Control Protocol)

- TCP offers connection oriented end-to-end packet delivery.
- TCP ensures reliability by sequencing bytes with a forwarding acknowledgement number that indicates to the destination the next byte the source expect to receive.
- It retransmits the bytes not acknowledged with in specified time period.

Web Protocol:

4. IP (Internet Protocol)

- The **Internet Protocol (IP)** is the foundational communication protocol that enables devices to **send and receive data across the internet** and other networks. It acts like a digital postal system, assigning **unique addresses (IP addresses)** to devices and ensuring data packets are correctly **routed** from source to destination.
- The **sender device** (IP: 192.16.00.12) initiates communication by encapsulating data into a packet, including both its **source IP address** and the **destination IP address**.
- The packet is then **forwarded to the internet**, which serves as the medium responsible for routing the packet based on the destination IP (192.00.00.75).
- **Routing protools and network infrastructure** interpret the destination IP and determine the most efficient path for delivering the packet.
- Upon reaching the destination, the **recipient device** (IP: 192.00.00.75) identifies the packet as intended for it and proceeds to process the received data.
- This process ensures **accurate and efficient data delivery** between networked devices, forming the core functionality of the Internet Protocol.

Web Protocol:

5. FTP (File Transfer Protocol)

- **FTP (File Transfer Protocol)**is a standard network protocol used to transfer files between a client and a server over the internet or a local network. It allows users to upload, download, delete, or manage files on a remote server.
- FTP is commonly used for website management, file sharing, and data backup. However, it does not encrypt data, making it less secure compared to modern alternatives like SFTP or FTPS.
- FTP is used to copy files from one host to another. FTP offers the mechanism for the same in following manner:
  - FTP creates two processes such as Control Process and Data Transfer Process at both ends i.e. at client as well as at server.
  - FTP establishes two different connections: one is for data transfer and other is for control information.
  - **Control connection** is made between **control processes** while **Data Connection** is made between<="" b="" style="box-sizing: border-box;">
  - FTP uses **port 21** for the control connection and **Port 20** for the data connection.

Web Protocol:

5. FTP (File Transfer Protocol)

- FTP uses two separate connections: the **Control Channel** and the **Data Channel**.
- The **Control Channel** is used to send commands (like login requests or file operation instructions), while the **Data Channel** handles the actual transfer of files (uploading or downloading).
- This separation allows efficient and organized file management over a network. A common use case is uploading website files to a server or downloading backups from it.

Web Protocol:

6. Trivial File Transfer Protocol (TFTP)

- **Trivial File Transfer Protocol** is also used to transfer the files but it transfers the files without authentication. Unlike FTP, TFTP does not separate control and data information.
- Since there is no authentication exists, TFTP lacks in security features therefore it is not recommended to use TFTP.
- **Key points**
  - TFTP makes use of UDP for data transport. Each TFTP message is carried in separate UDP datagram.
  - The first two bytes of a TFTP message specify the type of message.
  - The TFTP session is initiated when a TFTP client sends a request to upload or download a file.
  - The request is sent from an ephemeral UDP port to the **UDP port 69** of an TFTP server.

Web Protocol:

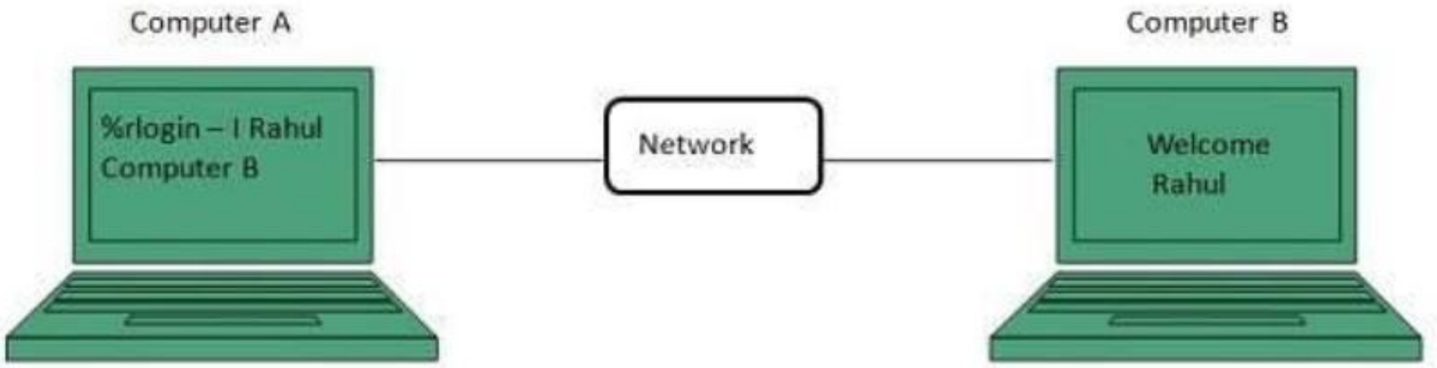
Difference between FTP and TFTP

S.N.	Parameter	FTP	TFTP
1	Operation	Transferring Files	Transferring Files
2	Authentication	Yes	No
3	Protocol	TCP	UDP
4	Ports	21 – Control, 20 – Data	Port 3214, 69, 4012
5	Control and Data	Separated	Separated
6	Data Transfer	Reliable	Unreliable

Web Protocol:

7. Telnet

- Telnet is a protocol used to log in to remote computer on the internet.
- There are a number of Telnet clients having user friendly user interface.
- The following diagram shows a person is logged in to computer A, and from there, he remote logged into computer B.



Web Browser

- **web Browser** is an application software that allows us to view and explore information on the web. User can request for any web page by just entering a URL into address bar.
- Web browser can show text, audio, video, animation and more. It is the responsibility of a web browser to interpret text and commands contained in the web page.
- Earlier the web browsers were text-based while now a days graphical-based or voice-based web browsers are also available. Following are the most common web browser available today:

Web Browser

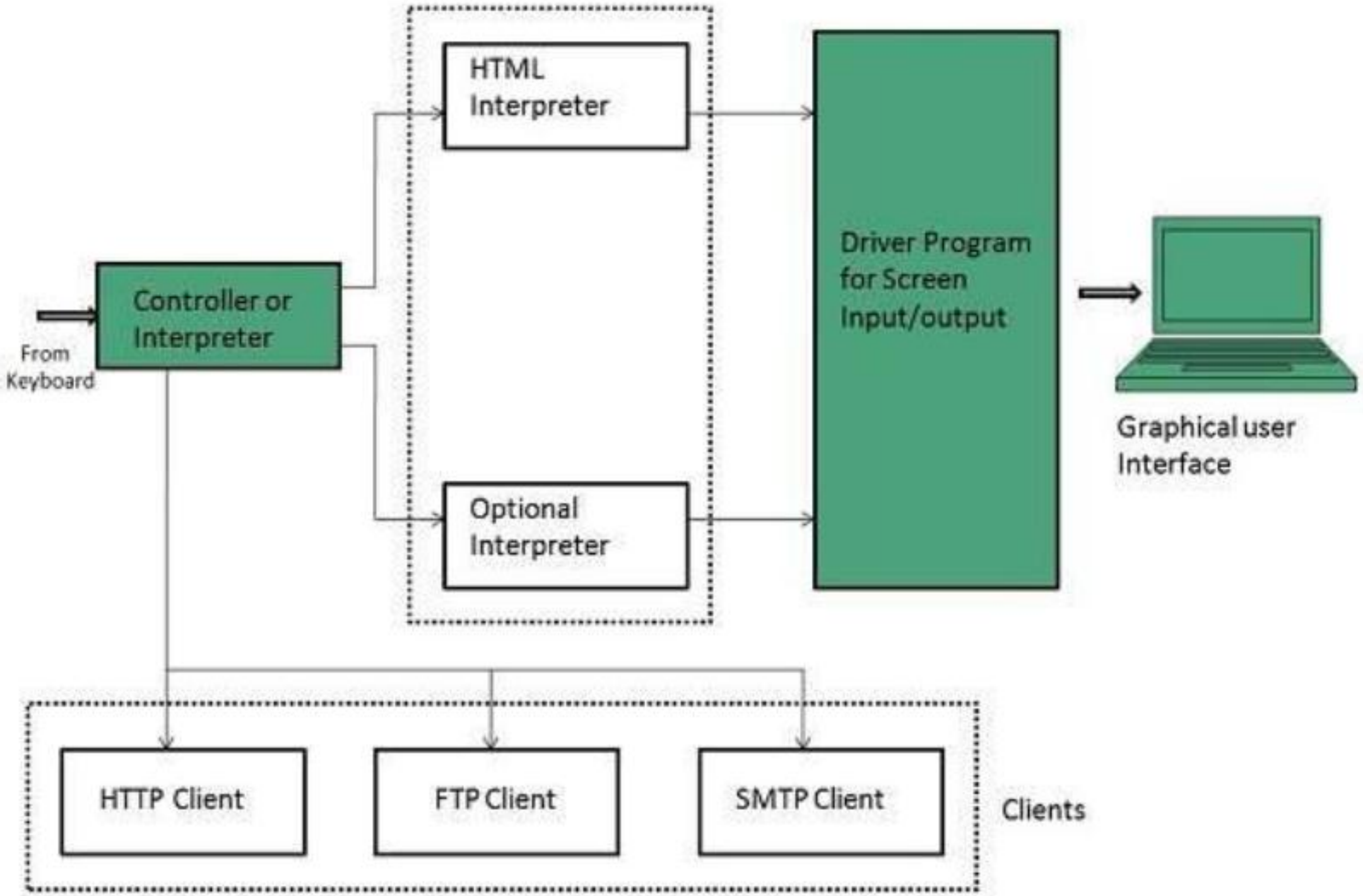
Browser	Vendor
Internet Explorer	Microsoft
Google Chrome	Google
Mozilla Firefox	Mozilla
Netscape Navigator	Netscape Communications Corp.
Opera	Opera Software
Safari	Apple
K-meleon	K-meleon

Architecture

- There are a lot of web browser available in the market. All of them interpret and display information on the screen however their capabilities and structure varies depending upon implementation. But the most basic component that all web browser must exhibit are listed below:
  - ✓ Controller/Dispatcher
  - ✓ Interpreter
  - ✓ Client Programs

- **Controller** works as a control unit in CPU. It takes input from the keyboard or mouse, interpret it and make other services to work on the basis of input it receives.
- **Interpreter** receives the information from the controller and execute the instruction line by line. Some interpreter are mandatory while some are optional For example, HTML interpreter program is mandatory and java interpreter is optional.
- **Client Program** describes the specific protocol that will be used to access a particular service. Following are the client programs that are commonly used: HTTP , SMTP, FTP , NNTP, POP

Architecture



Web server

- **Web server** is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc.
- Web site is collection of web pages while web server is a software that respond to the request for web resources.

Web Server Working:

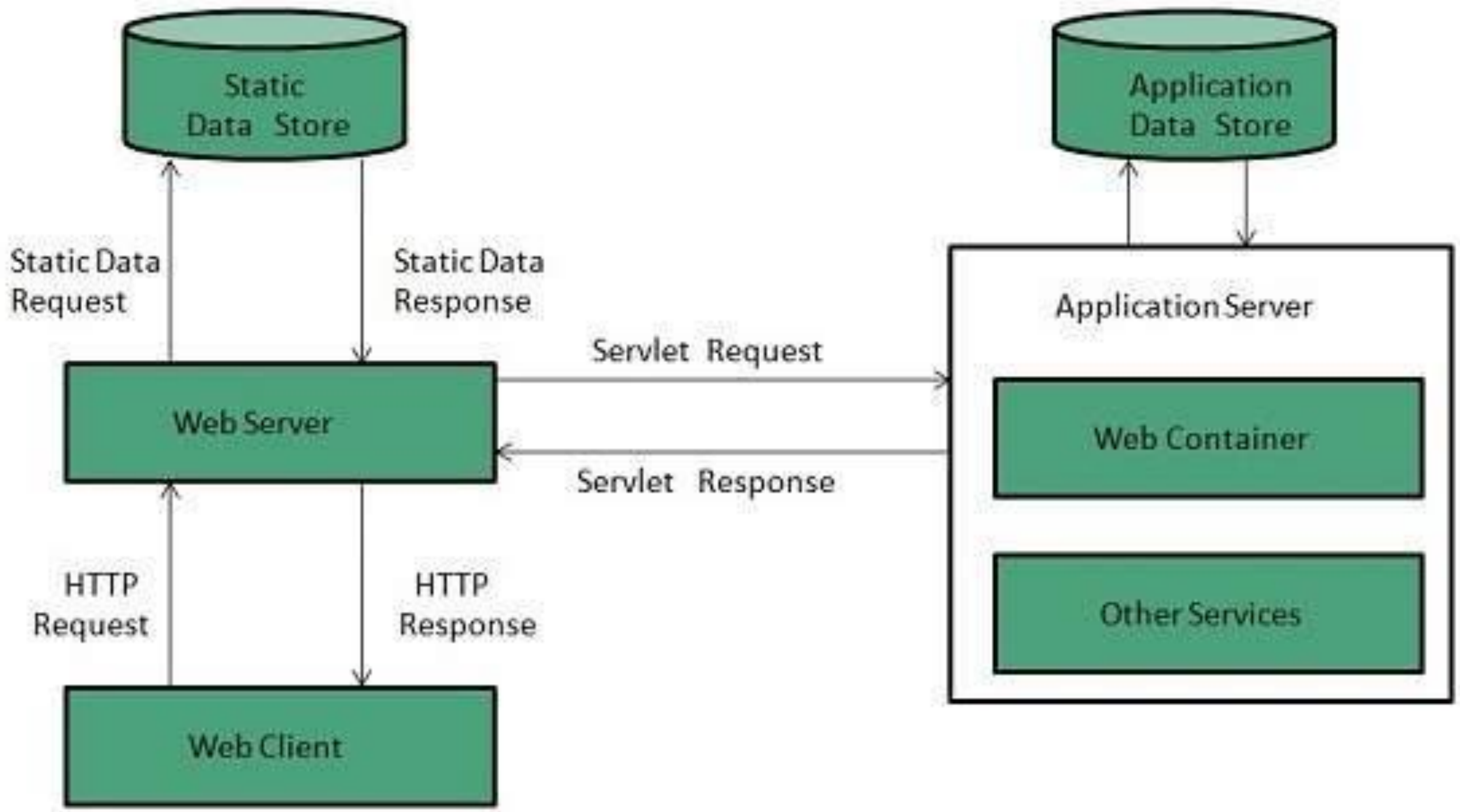
Web server respond to the client request in either of the following two ways:

- ✓Sending the file to the client associated with the requested URL.
- ✓Generating response by invoking a script and communicating with database

Web server

Key Points

- When client sends request for a web page, the web server search for the requested page if requested page is found then it will send it to client with an HTTP response.
- If the requested web page is not found, web server will the send an **HTTP response: Error 404 Not found.**
- If client has requested for some other resources, then the web server will contact to the application server and data store to construct the HTTP response.


Static
Data Store
Application
Data Store
Static Data
Request
Static Data
Response
Application Server
Servlet Request
Web Server
Web Container
Servlet Response
HTTP
Request
HTTP
Response
Other Services
Web Client

Web server Architecture

Web Server Architecture follows the following two approaches:

- ✓Concurrent Approach
- ✓Single-Process-Event-Driven Approach.

Concurrent Approach :

- Concurrent approach allows the web server to handle multiple client requests at the same time. It can be achieved by following methods:
  - ✓ Multi-process
  - ✓ Multi-threaded
  - ✓ Hybrid method.

Web server Architecture

Multi-threaded

- In this a single process (parent process) initiates several single-threaded child processes and distribute incoming requests to these child processes. Each of the child processes are responsible for handling single request.
- It is the responsibility of parent process to monitor the load and decide if processes should be killed or forked.

Multi-processing

- Unlike Multi-process, it creates multiple single-threaded process.

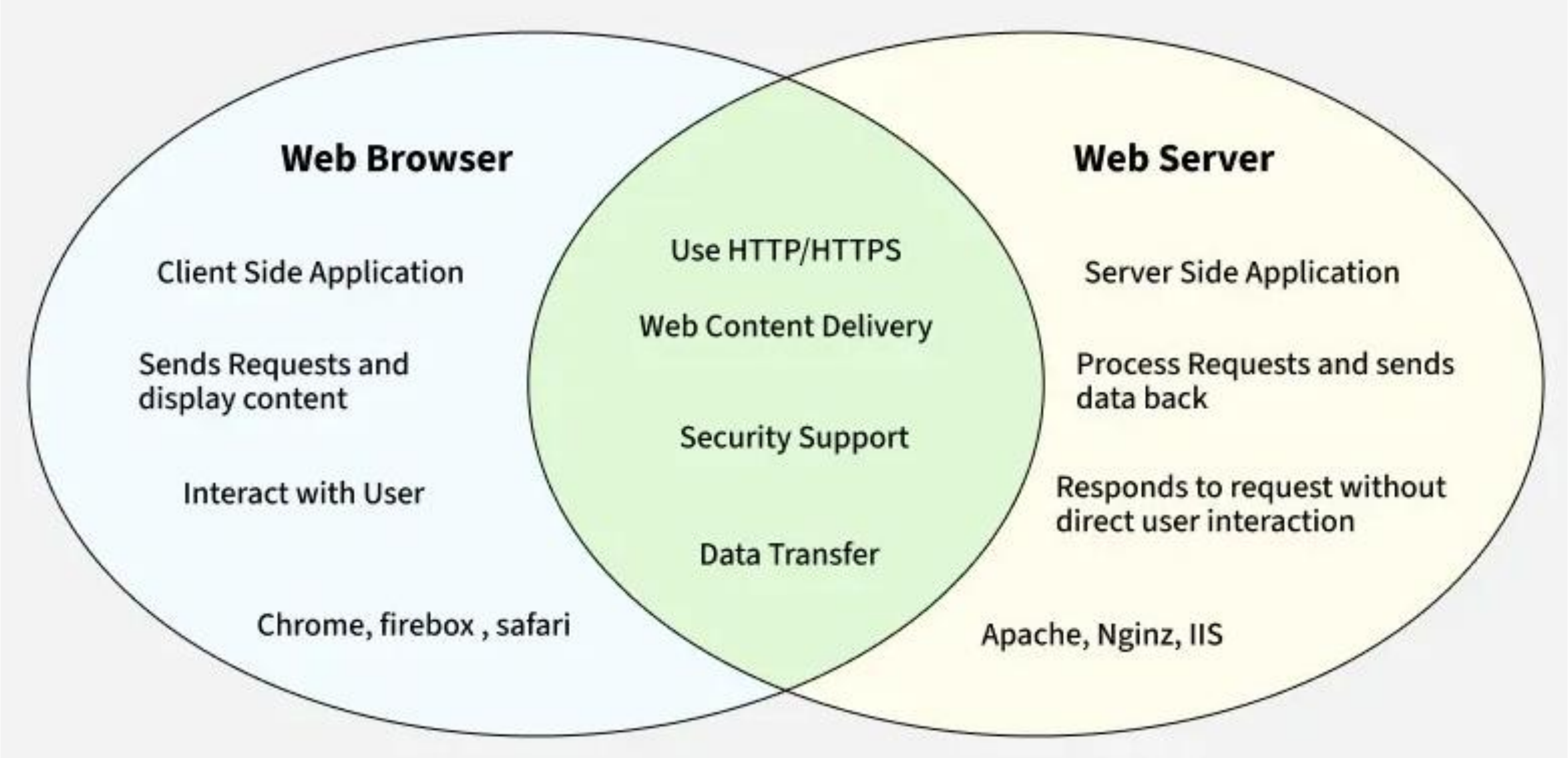
Hybrid

- It is combination of above two approaches. In this approach multiple process are created and each process initiates multiple threads. Each of the threads handles one connection. Using multiple threads in single process results in less load on system resources.

Web server Architecture

Multiprocessing	Multithreading
In Multiprocessing, CPUs are added for increasing computing power.	While In Multithreading, many threads are created of a single process for increasing computing power.
In Multiprocessing, Many processes are executed simultaneously.	While in multithreading, many threads of a process are executed simultaneously.
Multiprocessing are classified into **Symmetric** and **Asymmetric**.	While Multithreading is not classified in any categories.
In Multiprocessing, Process creation is a time-consuming process.	While in Multithreading, process creation is according to economical.
In Multiprocessing, every process owned a separate address space.	While in Multithreading, a common address space is shared by all the threads.

Web Browser	Web Server
A Web Browser is a client-side software used to access and view web pages.	A Web Server is a server-side software that stores and delivers web content.
Browsers are used by end-users to interact with websites and web applications.	Web servers handle incoming requests from browsers and send appropriate responses.
Popular browsers include Google Chrome, Firefox, Safari, and Edge.	Popular web servers include Apache, Nginx, and Microsoft IIS.
The browser sends HTTP/HTTPS requests to a server for content.	The server listens for incoming HTTP requests and sends data back to the browser.
A Web Browser can display HTML, CSS, JavaScript, images, and videos.	A Web Server hosts and serves web files (HTML, images, scripts) to browsers.
The Web Browser handles rendering the content that is returned by the server.	The Web Server handles storing, processing, and delivering content.
Browsers run on client devices like laptops, smartphones, and tablets.	Web Servers run on server machines or cloud infrastructure.
Browsers interpret data and display it in a user-friendly way.	Web Servers ensure the correct delivery of content over the Internet.
The browser can have additional features like bookmarks, extensions, and privacy settings.	Web Servers focus on efficient content delivery and request management.
The browser allows users to interact with the web (forms, clicks, etc.).	The server processes requests and returns data, but doesn’t interact with the user directly.
Browsers can be used for browsing, searching, and accessing web-based apps.	Web Servers are used for hosting websites, storing data, and processing requests.


Web Browser
Client Side Application
Sends Requests and display content
Interact with User
Chrome, firebox , safari
Use HTTP/HTTPS
Web Content Delivery
Security Support
Data Transfer
Web Server
Server Side Application
Process Requests and sends data back
Responds to request without direct user interaction
Apache, Nginz, IIS

Web Architecture: Client-Server Model

- The **Client-Server Model** is a distributed application architecture that divides tasks or workloads between **servers** (providers of resources or services) and **clients** (requesters of those services). In this model, a **client** sends a request to a **server** for data, which is typically processed on the server side. The server then returns the requested data to the client.
- **Clients** generally do not share resources with each other, but instead rely on the server to **provide the resources** or **services requested**. Common examples of the client-server model include email systems and the **World Wide Web** (WWW), where email clients interact with mail servers, and web browsers request resources from web servers.

Client-Server Model

How the Browser Interacts With the Servers?

The process of interacting with servers through a browser involves several steps. Here's a breakdown of the steps taken when you enter a URL in a browser and receive the website data:

1. User Enters the URL (Uniform Resource Locator): The user types a website address (e.g., www.example.com) into the browser's address bar.

2. DNS (Domain Name System) Lookup: The browser sends a request to the **DNS server** to resolve the human-readable URL into an IP address (since computers use IP addresses to identify and connect to each other).

3. DNS Server Resolves the Address: The DNS server looks up the domain name and returns the **IP address** of the web server hosting the requested website.

4. Browser Sends HTTP/HTTPS Request: The browser sends an **HTTP/HTTPS request** to the IP address of the web server to fetch the website’s data. HTTP (HyperText Transfer Protocol) or HTTPS (the secure version) is the protocol used for communication between the browser (client) and the web server (server).

Client-Server Model

5. Server Sends Website Files: The server processes the request and sends the necessary website files (HTML, CSS, JavaScript, images, etc.) back to the browser.

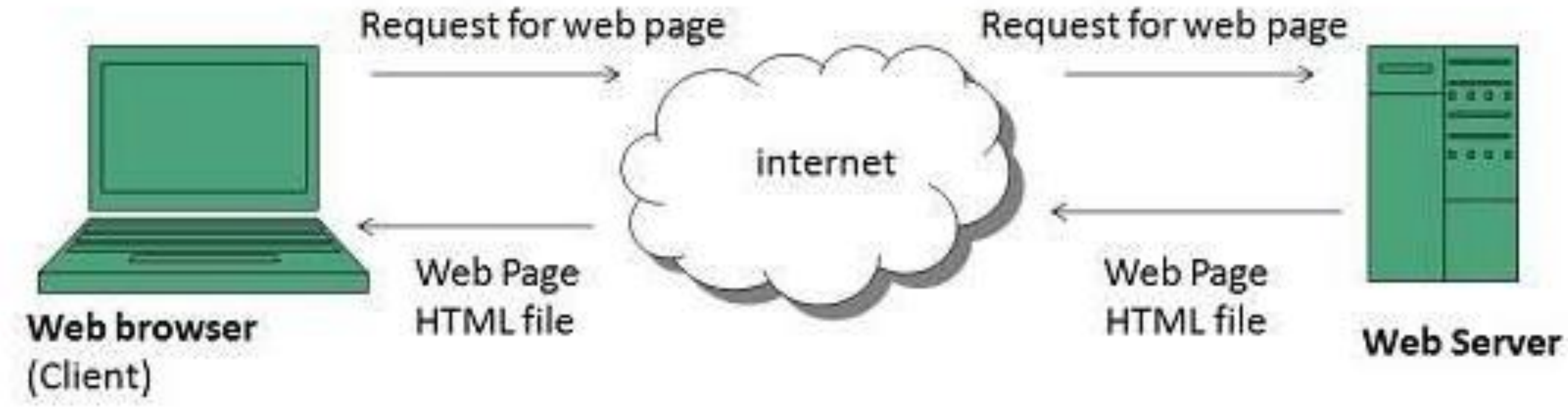
6. Rendering the Website: The browser **renders the files** and displays the website to the user. This rendering process involves several components:Together, these components, known as **Just-In-Time (JIT) Compilers**, allow the browser to convert raw data into a visual webpage.

DOM (Document Object Model) Interpreter: Processes the HTML structure.

CSS Interpreter: Applies styles to the HTML elements.

JS Engine: Executes JavaScript code for interactivity.

Together, these components, known as **Just-In-Time (JIT) Compilers**, allow the browser to convert raw data into a visual webpage.


Request for web page
Request for web page
internet
Web Page
HTML file
Web Page
HTML file
Web browser
(Client)
Web Server

Advantages of the Client-Server Model

- **The Client-Server model offers several advantages that make it popular in networked and distributed systems:**
- **Centralized Data Management**: All data is stored in a centralized server, which makes it easier to manage, update, and back up.
- **Cost Efficiency**: Since the server handles most of the processing, clients require fewer resources and can be simpler devices, reducing costs.
- **Scalability**: Both **clients** and **servers** can be scaled separately. Servers can be upgraded to handle more clients, and new clients can be added without significant changes to the server infrastructure.
- **Data Recovery**: Centralized data storage on the server allows for better **data recovery** and easier **backup** strategies.
- **Security**: **Security measures** such as firewalls, encryption, and authentication can be centralized on the server, ensuring that sensitive data is protected.

Disadvantages of Client-Server Model

- **Clients Are Vulnerable**: Clients are prone to viruses, Trojans, and worms if present in the Server or uploaded into the Server.
- **Servers Are Targets**: Servers are prone to Denial of Service (DOS) attacks, where the server is overwhelmed with traffic and made unavailable to legitimate clients.
- **Data Spoofing and Modification**: Data packets may be spoofed or modified during transmission if the proper security measures (e.g., encryption) are not implemented.
- **Man-in-the-Middle (MITM) Attacks**: Phishing or capturing login credentials or other useful information of the user are common and MITM(Man in the Middle) attacks are common.

Real-World Examples of the Client-Server Model

1. Email Systems

Client: The user's email client (e.g., **Microsoft Outlook**, **Gmail App**).

Server: The email server (e.g., **Gmail Server**, **Yahoo Mail Server**).

How It Works: The email client requests emails from the server, and the server delivers them. Similarly, when the user sends an email, the client communicates with the server to send the message.

2. The World Wide Web

Client: A web browser (e.g., **Google Chrome**, **Mozilla Firefox**).

Server: A web server (e.g., **Apache Server**, **Nginx Server**).

How It Works: The browser requests the web pages from the server, and the server sends the HTML files back to the client, which are then rendered and displayed.

3. Cloud Storage Services

Client: The user's device (e.g., smartphone, PC).

Server: A cloud server (e.g., **Google Drive**, **Dropbox**).

How It Works: The client uploads files to the server and can download them when needed. The server stores all the user's files centrally, ensuring **remote access** to data.

Web Page

- **web page** is a document available on world wide web. Web Pages are stored on web server and can be viewed using a web browser.
- A web page can contain huge information including text, graphics, audio, video and hyper links. These hyper links are the link to other web pages.
- Collection of linked web pages on a web server is known as **website.** There is unique **Uniform Resource Locator (URL)** is associated with each web page.
- There are two basic methods of web design: static and dynamic web pages. Users access static web pages, which present the same content every time they are viewed. On the other hand, dynamic webpages create content instantly in response to user input and present customized or updated information.

Static Web Page

- Static Web pages are very simple.
- It is written in languages such as HTML, JavaScript, CSS, etc. For static web pages when a server receives a request for a web page, then the server sends the response to the client without doing any additional process.
- These web pages are seen through a web browser. In static web pages, Pages will remain the same until someone changes it manually.

Scripting Languages

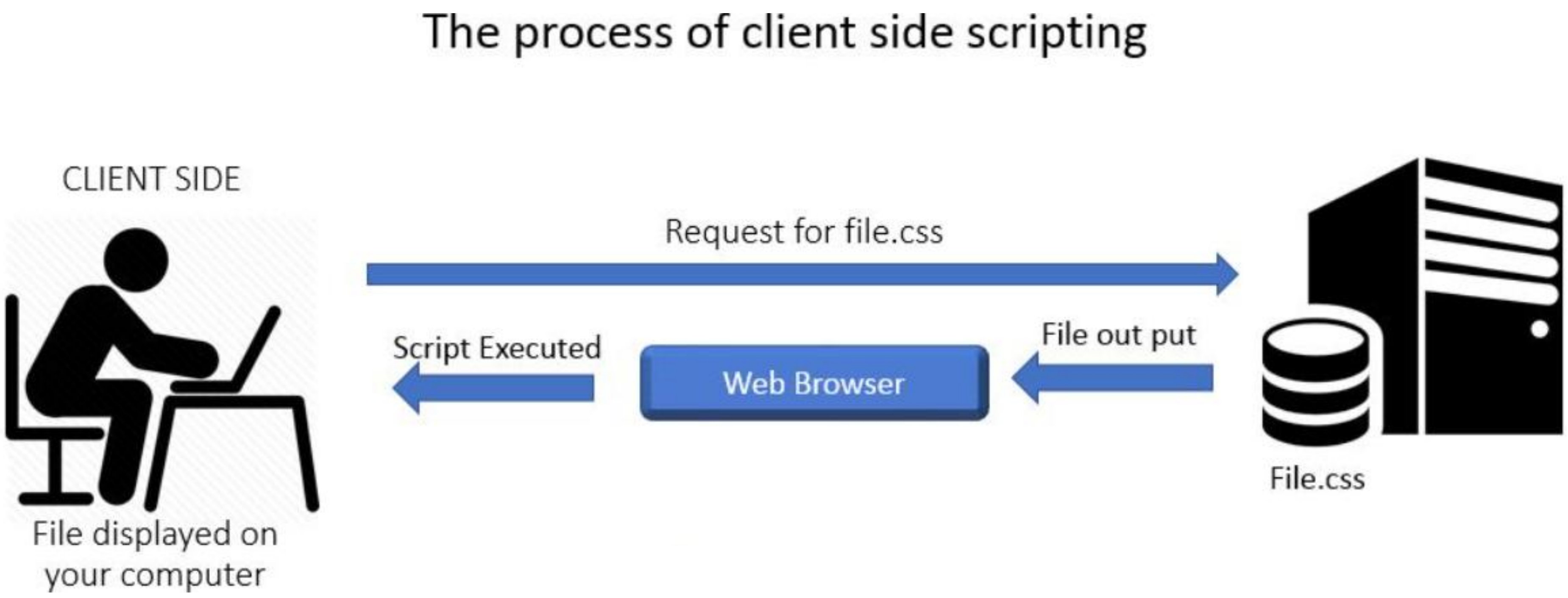
- Scripting languages are like programming languages that allow us to write programs in form of script. These scripts are interpreted not compiled and executed line by line.
- Scripting language is used to create dynamic web pages.

Client-side scripting

- Web browsers execute client-side scripting. It is used when browsers have all code.
- Source code is used to transfer from webserver to user's computer over the internet and run directly on browsers. It is also used for validations and functionality for user events.
- It allows for more interactivity.
- It usually performs several actions without going to the user. It cannot be basically used to connect to databases on a web server.
- These scripts cannot access the file system that resides in the web browser. Pages are altered on basis of the user's choice.
- It can also be used to create "cookies" that store data on the user's computer.

S.N.	Scripting Language Description
1.	**JavaScript** It is a prototype based scripting language. It inherits its naming conventions from java. All java script files are stored in file having **.js** extension.
2.	**ActionScript** It is an object oriented programming language used for the development of websites and software targeting Adobe flash player.
3.	**Dart** It is an open source web programming language developed by Google. It relies on source-to-source compiler to JavaScript.
4.	**VBScript** It is an open source web programming language developed by Microsoft. It is superset of JavaScript and adds optional static typing class-based object oriented programming.

Client-side scripting



Advantages of client-side scripting

1. More interactivity: Client-side scripting makes web pages more engaging for users.
2. Less server load: Client-side scripting reduces the server's tasks by handling validation and animation on the user's computer.
3. Better security: Client-side scripting protects sensitive data by keeping it on the user's machine instead of sending it to the server.
4. Instant feedback: Client-side scripting provides immediate responses to user actions, improving the website's responsiveness.
5. Improved scalability: Client-side scripting lightens the server's workload, allowing for better scalability.
6. Enhanced user experience: Client-side scripts enable dynamic content updates without page refreshing, creating a smoother and more enjoyable user experience

Disadvantages of client-side scripting

1. Security risks: Client-side scripts can be targeted by security attacks like cross-site scripting (XSS), exposing vulnerabilities.
2. Browser compatibility challenges: Developers need to ensure client-side scripts work across different web browsers, which can be a complex task.
3. Performance impact: Client-side scripts may cause slower page loading, especially on older or slower computers, affecting performance.
4. Restricted control: Client-side scripts have limited access to server-side resources, making complex operations and database interactions more challenging.

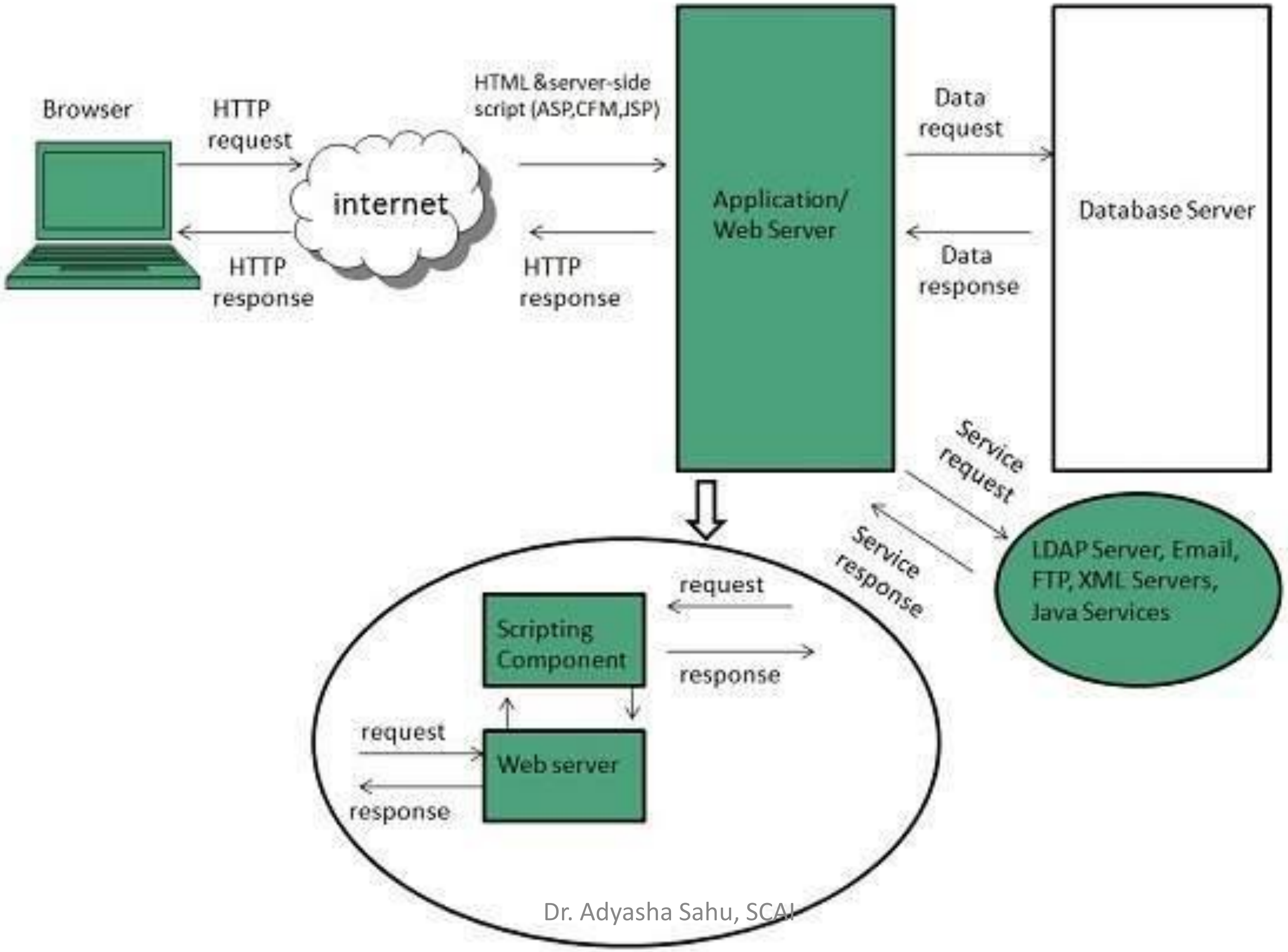
Example of client-side scripting

1. Validation: Client-side scripting can be used to validate form input, such as ensuring that a user enters a valid email address or phone number.
2. Animation: Client-side scripting can be used to add animation to web pages, such as a spinning logo or a scrolling banner.
3. Dynamic content updates: Client-side scripting can be used to update the content of a web page dynamically, such as displaying the current time or weather conditions.

Server-side scripting

- Web servers are used to execute server-side scripting. They are basically used to create dynamic pages.
- It can also access the file system residing at the webserver. A server-side environment that runs on a scripting language is a web server.
- Scripts can be written in any of a number of server-side scripting languages available.
- It is used to retrieve and generate content for dynamic pages.
- It is used to require to download plugins. In this load times are generally faster than client-side scripting.
- When you need to store and retrieve information a database will be used to contain data.
- It can use huge resources of the server. It reduces client-side computation overhead. The server sends pages to the request of the user/client.

Server-side scripting



Advantages of server-side scripting

1. Full control: Server-side scripts run on the server, which gives developers complete control over the environment.
2. Data security: Server-side scripts can access and process sensitive data on the server, which is more secure than storing it on the client's computer.
3. Scalability: Server-side scripts can be scaled to handle large numbers of requests.
4. Cross-browser compatibility: Server-side scripts generate HTML that is sent to the client, ensuring consistent behavior across different browsers.
5. Access to server resources: Server-side scripts have direct access to server resources like databases, file systems, and external APIs, enabling complex operations and data manipulation.

Disadvantages of server-side scripting

1. Increased server load: Server-side scripts can increase the load on the server, which can lead to slower performance.
2. Complexity: Server-side scripting can be more complex than client-side scripting, which can make it more difficult to develop and maintain.
3. Security: Server-side scripts are more vulnerable to security attacks than client-side scripts.
4. Limited interactivity: Server-side scripts generate HTML that is sent to the client, so they cannot provide immediate feedback or interact with the user without additional client-side scripting.

Example of server-side scripting

- Generating dynamic content: Server-side scripting can be used to generate dynamic content, such as search results, product listings, and user profiles.
- Processing data: Server-side scripting can be used to process data, such as calculating shipping costs or updating a database.
- Controlling access: Server-side scripting can be used to control access to web pages, such as requiring users to log in before they can view certain content.

S.N.	Scripting Language Description
1.	**ASP** Active Server Pages (ASP)is server-side script engine to create dynamic web pages. It supports **Component Object Model (COM)** which enables ASP web sites to access functionality of libraries such as DLL.
2.	**ActiveVFP** It is similar to PHP and also used for creating dynamic web pages. It uses native **Visual Foxpro** language and database.
3.	**ASP.net** It is used to develop dynamic websites, web applications, and web services.
4.	**Java** Java Server Pages are used for creating dynamic web applications. The Java code is compiled into byte code and run by **Java Virtual Machine (JVM).**
5.	**Python** It supports multiple programming paradigms such as object-oriented, and functional programming. It can also be used as non-scripting language using third party tools such as **Py2exe** or **Pyinstaller.**
6.	**WebDNA** It is also a server-side scripting language with an embedded database system.

Difference between client-side scripting and server-side scripting :

Client-side scripting	Server-side scripting
Source code is visible to the user.	Source code is not visible to the user because its output of server-sideside is an HTML page.
Its main function is to provide the requested output to the end user.	Its primary function is to manipulate and provide access to the respective database as per the request.
It usually depends on the browser and its version.	In this any server-side technology can be used and it does not depend on the client.
It runs on the user's computer.	It runs on the webserver.
There are many advantages linked with this like faster. response times, a more interactive application.	The primary advantage is its ability to highly customize, response requirements, access rights based on user.
It does not provide security for data.	It provides more security for data.

Difference between client-side scripting and server-side scripting :

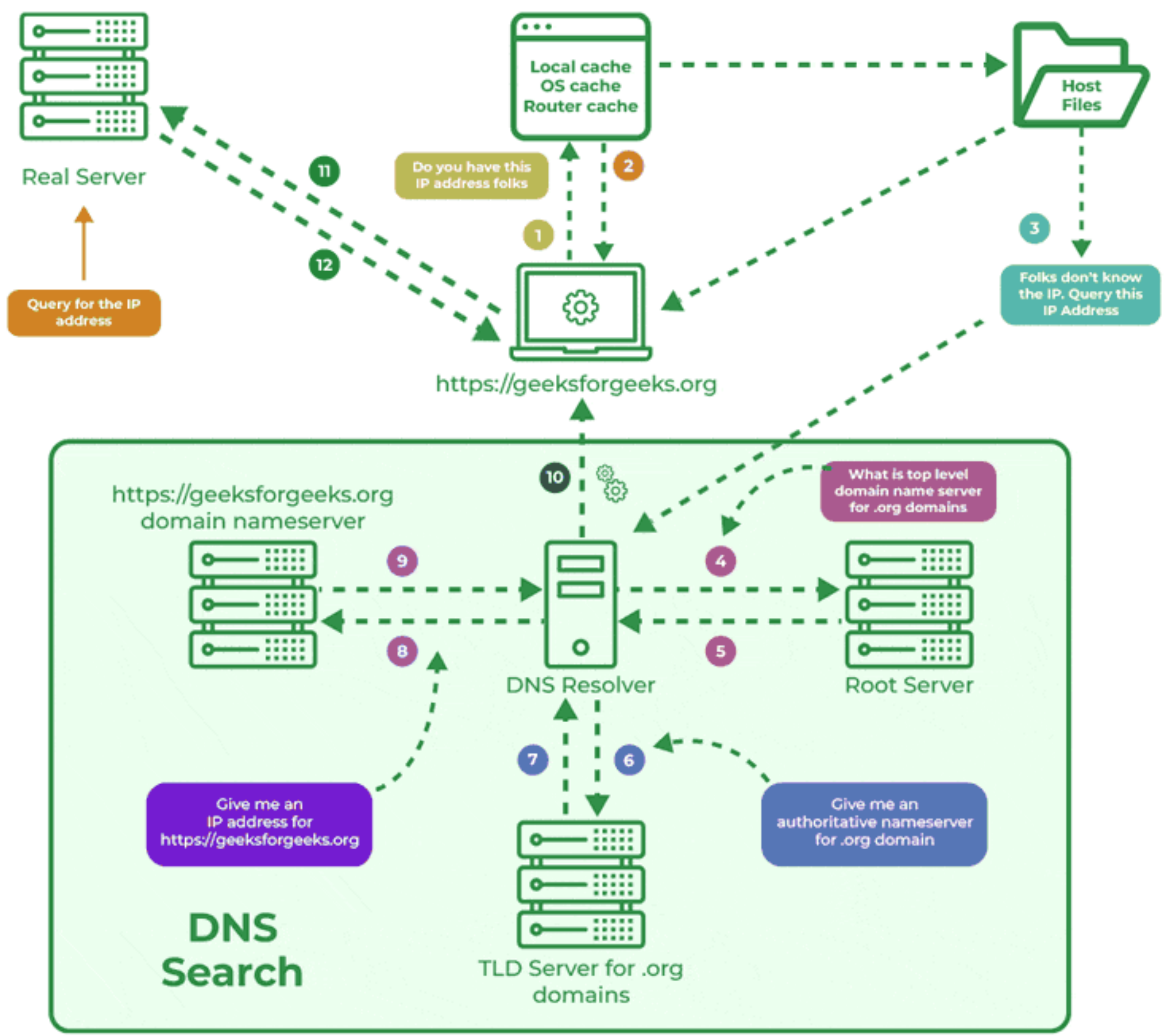
Client-side scripting	Server-side scripting
It is a technique used in web development in which scripts run on the client's browser.	It is a technique that uses scripts on the webserver to produce a response that is customized for each client's request.
HTML, CSS, and javascript are used.	PHP, Python, Java, Ruby are used.
No need of interaction with the server.	It is all about interacting with the servers.
It reduces load on processing unit of the server.	It surge the processing load on the server.

Parameters	Server Side Scripting	Client-Side Scripting
Script Running	For server-side scripting, a web server serves as the medium for running it. They create the pages that one would send to the browser.	The script for the client-side scripting runs by using a browser. It is present already in any user’s computer.
Uses	We use the server-side scripting at the back-end, where the source code stays hidden from the browser (client-side)- making it non-viewable.	We use the client-side scripting at the front end, and any user can view it using the browser itself.
Occurrence	Server-side scripting occurs whenever a browser initiates a request for it. As a result, many dynamic pages get created on the basis of several conditions.	Client-side scripting occurs when all the codes that a browser possesses in a page later change/ alter according to any user’s input.
Operation	Any server is capable of carrying out server-side scripting, but it can’t carry out client-side scripting.	A typical browser performs the client-side scripting after it receives a page (that the server sends).
Execution	The server-side scripting occurs on a remote computer. Thus, the response we get is slower as compared to that of the client-side scripting.	The client-side scripting occurs on the local computers. Thus, in this case, we get a comparatively quicker response in comparison with the server-side scripting.
Suitability	The server-side scripting works well for the areas that require the loading of a dynamic type of data and information.	The client-side scripting works well in those cases that require user interaction.
Connection to Server Database	The server-side scripting assists a user in connecting to the database that already exists in the concerned web server.	The client-side server doesn’t connect to those databases that exist primarily on the concerned web server.
Access To Various Files	The server-side scripting has complete access to all the files present in any web server.	The client-side scripting has no access to the files that exist in a web server.
Languages	Languages like Ruby on Rails, Perl, ASP, Python, ColdFusion, PHP, etc., come into play in the case of server-side scripting.	Languages like VB Script, CSS, HTML, Javascript, etc., are very common in the case of client-side scripting.
Security	The server-side scripting is way more secure as compared to the client-side one. It is because the scripts of the server-side stay hidden from any random clients.	The client-side scripts are much less secure as compared to the server-side ones. It is because these scripts don’t stay hidden from any random client’s end.

Internet Domain Name System

- When **DNS** was not into existence, one had to download a **Host file** containing host names and their corresponding IP address. But with increase in number of hosts of internet, the size of host file also increased. This resulted in increased traffic on downloading this file. To solve this problem the DNS system was introduced.
- **Domain Name System** helps to resolve the host name to an address. It uses a hierarchical naming scheme and distributed database of IP addresses and associated names

How Does DNS Works



DNS

1) **User Input:** You enter a website address (for example, www.geeksforgeeks.org) into your web browser.
2) **Local Cache Check:** Your browser first checks its local cache to see if it has recently looked up the domain. If it finds the corresponding IP address, it uses that directly without querying external servers.
3) **DNS Resolver Query:** If the IP address isn't in the local cache, your computer sends a request to a DNS resolver. The resolver is typically provided by your Internet Service Provider (ISP) or your network settings.
4) **Root DNS Server:** The resolver sends the request to a root DNS server. The root server doesn't know the exact IP address for www.geeksforgeeks.org but knows which Top-Level Domain (TLD) server to query based on the domain's extension (e.g., .org).
5) **TLD Server:** The TLD server for .org directs the resolver to the authoritative DNS server for geeksforgeeks.org.
6) **Authoritative DNS Server:** This server holds the actual DNS records for geeksforgeeks.org, including the IP address of the website's server. It sends this IP address back to the resolver.
7) **Final Response:** The DNS resolver sends the IP address to your computer, allowing it to connect to the website's server and load the page.

DNS

Structure of DNS

- DNS operates through a hierarchical structure, ensuring scalability and reliability across the global internet infrastructure. Here's how it’s organized:
- **Root DNS Servers:** These are the highest-level DNS servers and know where to find the TLD servers. They are crucial for directing DNS queries to the correct locations.
- **TLD Servers:** These servers manage domain extensions like .com, .org, .net, .edu, .gov and others. They help route queries to the authoritative DNS servers for specific domains.
- **Authoritative DNS Servers:** These are the servers that store the actual DNS records for domain names. They are responsible for providing the correct IP addresses that allow users to reach websites.

IP Address

- IP address is a unique logical address assigned to a machine over the network. An IP address exhibits the following properties:
  - ✓ IP address is the unique address assigned to each host present on Internet.
  - ✓ IP address is 32 bits (4 bytes) long.
  - ✓ IP address consists of two components: **network component** and **host component**.
  - ✓ Each of the 4 bytes is represented by a number from 0 to 255, separated with dots. For example 137.170.4.124
- IP address is 32-bit number while on the other hand domain names are easy to remember names. For example, when we enter an email address we always enter a symbolic string such as webmaster@tutorialspoint.com.

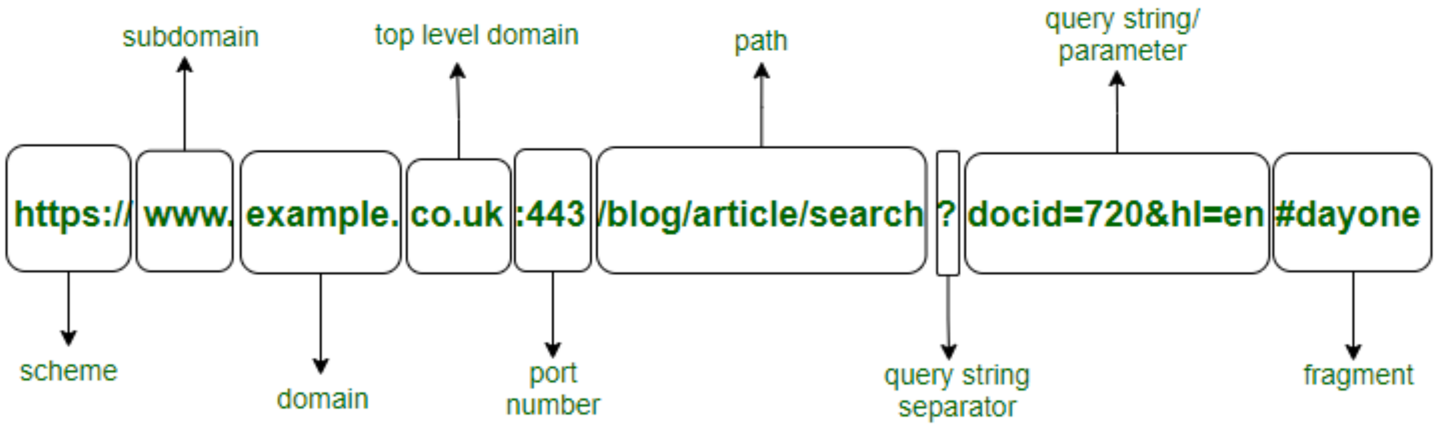
Uniform Resource Locator (URL)

- **Uniform Resource Locator (URL)** refers to a web address which uniquely identifies a document over the internet.
- This document can be a web page, image, audio, video or anything else present on the web.
- For example, **www.tutorialspoint.com/internet_technology/index.html** is an URL to the index.html which is stored on tutorialspoint web server under internet_technology directory.
- **URL Types** : There are two forms of URL as listed below:
  - ✓ Absolute URL
  - ✓ Relative URL

Uniform Resource Locator (URL) Structure

Parts of a URL

URL : https://www.example.co.uk:443/blog/article/search?docid=720&hl=en#dayone



Uniform Resource Locator (URL) Structure

- **Protocol:** The protocol or scheme part of the URL and indicates the **set of rules that will decide the transmission and exchange of data**. HTTPS which stands for Hyper Text Transfer Protocol Secure tells the browser to **display the page in Hyper Text (HTML) format** as well as **encrypt any information** that the user enters in the page. Other protocols include the **FTP** or File Transfer Protocol which is used for transferring files between client and server, **SMTP** or Single Mail Transfer Protocol which is used for sending emails.
- **Subdomain:** The subdomain is used to separate different sections of the website as it **specifies the type of resource** to be delivered to the client. Here the subdomain used 'www' is a general symbol for any resource on the web. Subdomains like 'blog' direct to a blog page, 'audio' indicates the resource type as audio.
- **Domain name: It specifies the organization or entity** that the URL belongs to. Like in *www.facebook.com* the domain name 'facebook' indicates the organization that owns the site.
- **Top-level domain:** The TLD (top-level domain) **indicates the type of organization the website is registered to**. Like the **.com** in *www.facebook.com* indicates a **commercial entity**. Similarly, .org indicates organization, .co.uk a commercial entity in the UK.

Uniform Resource Locator (URL) Structure

- **Port Numbrer:** A port number specifies **the type of service** that is requested by the client since **servers often deliver multiple services**. Some default port numbers include 80 for HTTP and 443 for HTTPS servers.
- **Path:** Path **specifies the exact location** of the web page, file, or any **resource that the user wants access to**. Like here the path indicates a specific article in the blog webpage.
- **Query String Separator:** The **query string** which contains specific parameters of the search is **preceded by a question mark (?).** The question mark tells the browser that **a specific query is being performed**.
- **Query String:** The query string **specifies the parameters of the data that is being queried from a website's database.** Each query string is **made up of a parameter and a value** joined by the equals (=) sign. In case of multiple parameters, query strings are joined using the ampersand (&) sign. The parameter can be a number, string, encrypted value, or any other form of data on the database.
- **Fragment:** The fragment identifier of a URL is **optional**, usually appears at the end, and begins with a hash (#). It indicates a **specific location within a page such as the 'id' or 'name' attribute for an HTML element**.

Uniform Resource Locator (URL) Structure

- **Domain or Host Name:**
  It is the reference or name of the page that you are going to access on the internet. In this case, the domain name is: **www.students.org.**
- **Port Name:**

It is defined just after the domain name by using the colons between itself and the domain name. Generally, it is not visible in the URL. The **domain name** and the **port name** combinely can be known as **Authority**. The default port for web services is **port80 (:80).**

Path:

It refers to the path or location of a particular file or page stored on the web server to access the content of it. The path used here is: **array-data-structure.**

Query:

A query mainly found in the dynamic pages. It consists of a **question mark(?)** followed by the parameters. In above URL query is: **?.**

Parameters:

These are the pieces of information inside a query string of URL. Multiple parameters can be passed to a URL by using the **ampersand(&)** symbol to separate them. The query parameter in above URL is: **ref=home-articlecards.**

Fragments:

The fragments appear at the end of a URL starts with a **Hashtag(#)** symbol. These are the internal page references that refers to a specific section within the page. The fragment in the above URL is: **#what-is-array**.

Absolute URL

- Absolute URL is a complete address of a resource on the web. This completed address comprises of protocol used, server name, path name and file name.
- For example http:// www.tutorialspoint.com / internet_technology /index.htm. where:
  - ✓ **http** is the protocol.
  - ✓ **tutorialspoint.com** is the server name.
  - ✓ **index.htm** is the file name.

- The protocol part tells the web browser how to handle the file. Similarly we have some other protocols also that can be used to create URL are:
  - ✓ FTP
  - ✓ https
  - ✓ Gopher
  - ✓ mailto
  - ✓ news

Relative URL

- Relative URL is a partial address of a webpage. Unlike absolute URL, the protocol and server part are omitted from relative URL.
- Relative URLs are used for internal links i.e. to create links to file that are part of same website as the WebPages on which you are placing the link.
- For example, to link an image on tutorialspoint.com/internet_technology/internet_referemce_models, we can use the relative URL which can take the form like **/internet_technologies/internet-osi_model.jpg.**

Difference between Absolute and Relative URL

Absolute URL	Relative URL
Used to link web pages on different websites	Used to link web pages within the same website.
Difficult to manage.	Easy to Manage
Changes when the server name or directory name changes	Remains same even of we change the server name or directory name.
Take time to access	Comparatively faster to access.